FootFallCam technical assessment

Overview

I built a video pipeline to find people wearing the staff badge and show when and where they appear in the video.

Pipeline and tool choices

OpenCV reads the video frame by frame and handles the image processing and annotated video. I chose pretrained YOLO11s to detect people so I could start with a model that already recognises people, without needing to collect and label a new person dataset. As the camera looks down on the scene, I run detection on four rotated versions of each frame and map the boxes back. Non-maximum suppression removes duplicate boxes. For this fixed camera, background subtraction can also propose moving regions that YOLO may have missed.

Tracking and badge detection

I use ByteTrack to connect person detections across frames and give each track an ID. This lets the system gather badge evidence over time, even when the badge is briefly hidden. I chose template matching for the badge because the assessment provided reference images. The program searches each person crop against rotated and resized versions of those images, every second frame. This keeps the search focused on the person and reduces processing.

Staff decision and outputs

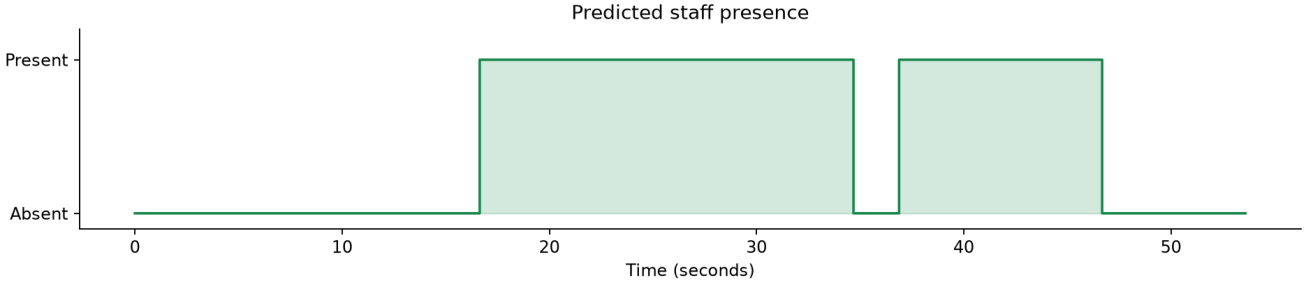
A track is marked as staff after at least two badge matches score 0.85 or higher within two seconds. I use more than one match to reduce decisions based on a single possible match. The label then carries across that track, including brief moments when the badge cannot be seen. Short gaps in the person boxes are filled by interpolation and marked as estimates. The program saves an annotated video, per-frame CSV results, a staff-presence timeline, and a plot of staff positions. Positions are measured in image pixels, with (0, 0) at the top left.

Results

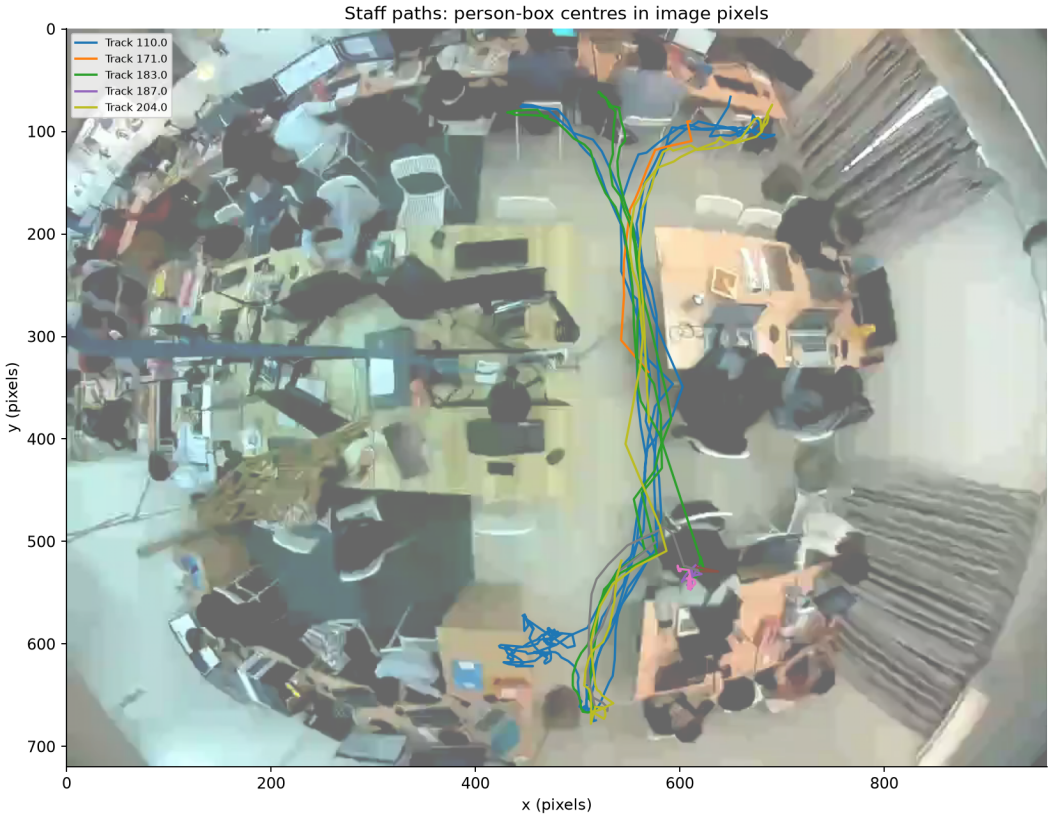
The program processed all 1,341 frames of the sample video, which is 960 by 720 pixels and runs at 25 frames per second. It marked 696 frames as containing staff across five track segments. Processing took about 6 minutes 17 seconds, at roughly 3.73 frames per second. Reviewing the output showed some missed staff intervals. Accuracy has not been measured against human-labelled frames, so these results describe the run on this sample only.

Visual outputs

Staff presence over time



Staff position in the video



The lines show the tracked staff positions in image pixels.